

Technique 4: Symmetry

MP

Teach You Physics Hand In Hand

May 2026

1 SOLUTION to Problem in Hollow Object Post

- Level of difficulty: **2/10**.
- Assume the hollow circle has a negative mass and superimposed onto the disc.

By area ratio, the mass of the hollow circle m is:

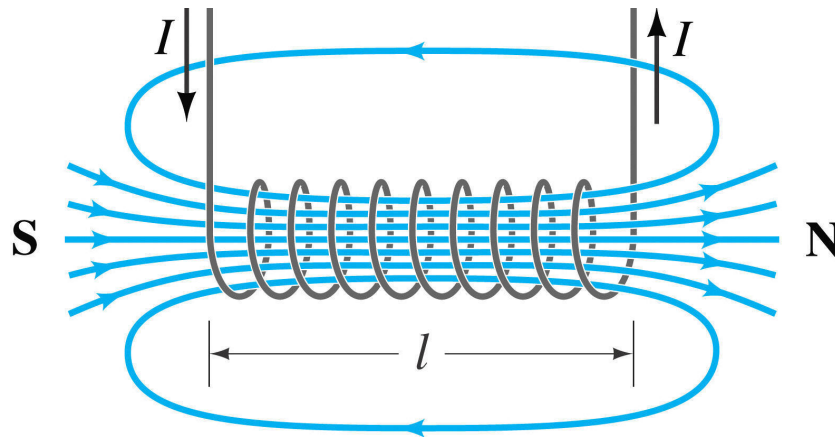
$$\frac{M}{|m|} = \frac{1}{(\frac{1}{2})^2} \Rightarrow m = -\frac{M}{4}$$

Center of mass from the left edge:

$$x_{CM} = \frac{MR + (-\frac{M}{4})(\frac{3R}{2})}{M + (-\frac{M}{4})} = \frac{5R}{6} \Rightarrow \text{Coordinates of the center of mass : } x_{CM} : (-\frac{R}{6}, 0)$$

2 Equal contribution = symmetry

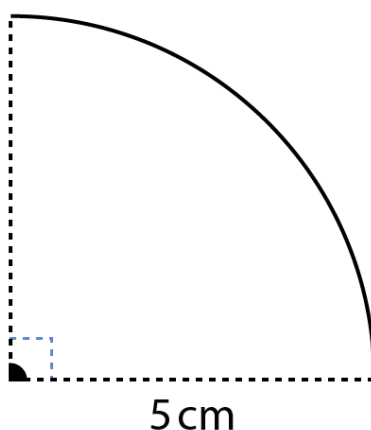
Consider a long solenoid with coil density n in coordinate plane. The magnetic field inside the solenoid is simply $B = \mu_0 n I$.



But what are the magnetic fields right at the edges? Here, we can consider symmetry. Imagine the solenoid is cut in the middle and separated into 2 long solenoids. **Since their contributions to the magnetic field strength at the point of cut are the same**, the magnetic field at the point of cut, which is the edge, is $B' = \frac{1}{2}\mu_0 nI$.

3 Example

Question: Consider a charged wire made into a shape of quarter circle with radius 5 cm.



The total amount of charge is 10 C. Find the electric field strength at the center.

Answer: By symmetry, the horizontal component is half of that created by a semi-circle. We have:

$$E_x = \frac{1}{2} \int \frac{\lambda dl}{4\pi\epsilon_0 R^2} \cos \theta = \frac{1}{2} \int_0^{2R} \frac{\lambda dy}{4\pi\epsilon_0 R^2} = \frac{1}{2} \frac{\lambda(2R)}{4\pi\epsilon_0 R^2} = \frac{Q}{2\pi^2\epsilon_0 R^2}$$

$$\Rightarrow E = \frac{Q}{\sqrt{2}\pi^2\epsilon_0 R^2} = 3.24 \times 10^{13} \text{ NC}^{-1}$$

When we see more complicated problems, considering symmetry becomes a short cut.

4 Problem

[20 points] Consider a tetrahedral resistance network. Each side of the tetrahedron has a resistance R . A current is flowing from A to B as shown.

a) [15 points] By the method of considering symmetry, find the equivalent resistance of the network.

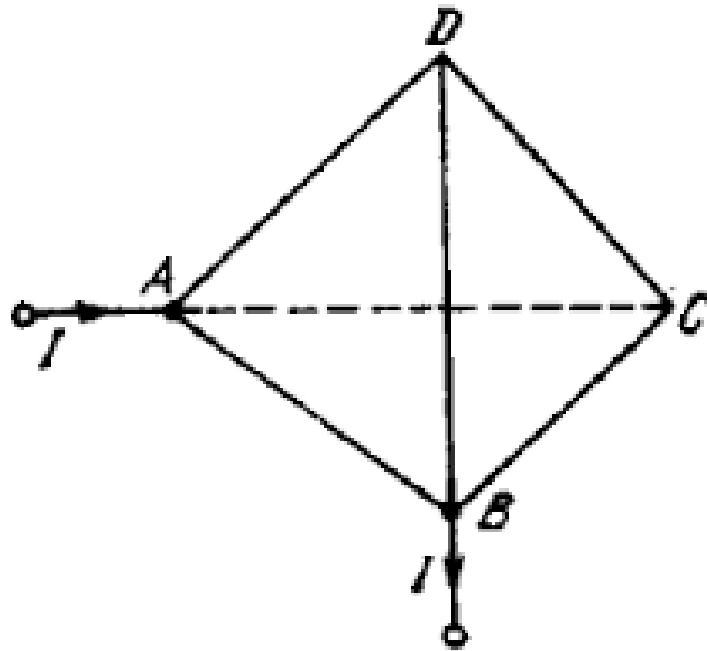


Fig. 82

b) [5 points] Is there any current flowing through CD?

The solution will be uploaded in the next post and MP love you all :)